Linear Topology

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Undefined Terms: Point; Precedes

x precedes y : $x < y$

Axiom I: S is a collection of points such that

(a) if x and y are distinct points, then either $x < y$ or $y < x$.

(b) if $x < y$, then x is not y.

(c) if $x < y$ and $y < z$, then $x < z$.

Theorem 1: If $x < y$, then y does not precede x.

Definition: If x is a point of the set M such that no point of M precedes x, then x is called the *first point* of M.

Define *last point* similarly.

Theorem 2: No set has two first points.

Theorem 3: Every finite set has a first point and a last point.

Theorem 4: If M is a set consisting of n points, then these points can be named $a_1, a_2, \cdots a_n$ so that $a_1 < a_2 < a_3 \cdots < a_n$.

Definition: If $x < z$ and $z < y$, then z is said to be *between* x and y. The set of points between x and y is called a *region* and will be referred to as the region xy.

Axiom II: S has no first point and no last point.

Theorem 5: If x is a point, then there is a region containing x.

Question: Is there a space differing from the x-axis in something other than size, that satisfies Axioms I and II? Ex: $\{n|$ n is an integer, $\{[0, z)|z \geq 0\}$

Definition: A point p is said to be a *limit point* of the set M, provided each region containing p contains a point of M distinct from p.

Theorem 6: If the set H is a subset of the set K and p is a limit point of H, then p is a limit point of K.

Theorem 7: If the point x is common to the regions R_1 and R_2, then some region is the common part of R_1 and R_2 and contains x.

Theorem 8: If the point p is a limit point of the sum of the two sets H and K, then p is a limit point of at least one of the sets H and K.

Theorem 9: If the point p is a limit point of the sum of a finite number of point sets, then p is a limit point of at least one of these sets.

Definitions: Two point sets are said to be *mutually exclusive* if they have no point in common.

If G is a collection of point sets such that each two of them are mutually exclusive, then the sets of G are said to be mutually exclusive.

Theorem 10: If x and y are two points, then there exist two mutually exclusive regions, one containing x and the other containing y.

Theorem 11: No finite point set has a limit point.

Theorem 12: If p is a limit point of the point set M, then every region containing p contains infinitely many points of M.

Definition: A sequence of points $p_1, p_2, p_3, \cdots$ is said to *converge* to a point p if for any region R containing p, there is a positive integer k such that if $n > k$, pn lies in R.

Theorem 13: No sequence of points converges to two points.

Theorem 14: If a is a sequence of points converging to a point p, then every subsequence of a converges to p.

Theorem 14.5: If α is a sequence of distinct points which converges to a point p, and β is a sequence of distinct points such that each point in β is a point in α, then β converges to p.

Theorem 15: If $p_1, p_2, p_3, \cdots$ is a sequence of distinct points converging to the point p, then p is the only limit point of the set $p_1 + p_2 + p_3 + \cdots$.

Theorem 16: If $p_1, p_2, p_3, \cdots$ is a sequence of points converging to a point p, then some region contains the set $p + p_1 + p_2 + p_3 + \cdots$.

Definition: A set is said to be *closed* if it contains all its limit points.

Notation: If H is a set, $\overline{H}$ denotes its closure (i.e. H plus all limit points of H).

Definition: A set H is said to be *open* if for each point x in H, there is a region containing x and lying in H.

Theorem 17: If H is a set, then $\overline{H}$ is closed.

Theorem 18: If x is a point, then the set of all points which precede x is an open set.

Theorem 19: Every region is an open set.

Definition: If x and y are two points, the set consisting of x, y and all points between x and y is called the *interval* xy.

Theorem 20: Every interval is closed.

Theorem 21: If the open set M is a proper subset of S, then $S \notin M$ (points of S which are not in M) is closed.

Theorem 22: If the closed set M is a proper subset of S, then $S \notin M$ is open.

Definition: Let G be a collection of sets. The set consisting of all points which are contained in at least one set of G is called $G*$.

Theorem 23: If G is a finite collection of closed sets, then $G*$ is closed.

Theorem 24: If G is a collection of open sets, then $G*$ is open.

Theorem 25: If G is a collection of closed sets having a common point, then $\cap G$ (the set of all points common to all sets of G) is closed.

Theorem 26: If G is a finite collection of open sets, then $\cap G$ is open.

Problem: Give examples to show that Theorems 23 and 26 are false if the word 'finite' is omitted.

Definition: Two sets are said to be *mutually separated* if they have no point in common and neither of them contains a limit point of the other.

Definition: A point set is said to be *connected* if it is not the sum of two mutually separated point sets.

Theorem 27: If H and K are two mutually separated point sets and M is a connected subset of $H + K$, then M is a subset of one of the sets H and K.

Theorem 28: Every point is a connected set.

Theorem 29: If a finite point set M contains more than one point, then M is not connected.

Theorem 30: If G is a collection of connected point sets having a common point, then $G*$ is connected.

Theorem 30.5: If G is a collection of connected sets having a common point, then $\cap G$ is connected.

Theorem 31: If M is a connected point set, then $\overline{M}$ is connected.

Definition: A point set is said to be *nondegenerate* if it contains more than one point.

Theorem 32: If M is a nondegenerate connected point set, then every point of M is a limit point of M.

Theorem 33: If x and y are two points of a connected point set M, then the region xy lies in M.

Axiom III: S is connected.

Theorem 34: If H and K are two sets such that $H \cup K = S$, and every point of H precedes every point of K, then either H has a last point or K has a first point, but not both.

Theorem 35: Every interval is connected.

Theorem 36: Every region is connected.

Theorem 37: No proper subset of S is both open and closed.

Theorem 38: If x and y are two points, then there is a point between them.

Question: Does Theorem 34 => Axiom III? Same question for Theorems 35, 36, 37, and 38.

Theorem 39: Every point of a region R is a limit point of R.

Theorem 40: Every point of an interval I is a limit point of I.

Theorem 41: No region has a first point.

Theorem 42: S contains infinitely many points, and each point of S is a limit point of S.

Definition: A sequence of points is said to be *bounded* if some region contains every point of this sequence.

Definition: A sequence of points $p_1, p_2, p_3, \cdots$ is said to be an *increasing sequence* if for each positive integer n, $p_n < p_{n+1}$.

Theorem 43: Every bounded increasing sequence of points converges.

Theorem 44: If a is a bounded sequence of points, then some subsequence of a converges.

Definition: A point set is said to be *bounded* if it lies in some region.

Theorem 45: Every bounded infinite point set has a limit point.

Theorem 46: Every closed and bounded point set contains a first point and a last point.

Theorem 47: If $M_1, M_2, M_3 \cdots$ is a sequence of closed and bounded point sets such that for each n, M_n contains M_{n+1}, then the sets $M_1, M_2, M_3 \cdots$ have a point in common. Furthermore, the set of all points common to these sets is closed.

Definition: A collection G of point sets is said to cover a point set M if every point of M lies in at least one set of G.

Theorem 48: If ab is a region and G is a collection of regions covering $\overline{ab}$, then some finite subcollection of G covers $\overline{ab}$.

Theorem 49: Strengthen Theorem 48 by letting G be a collection of open sets.

Theorem 50: If G is a collection of open sets covering the closed and bounded point set M, then some finite subcollection of G covers M.

Definition: A point set M is said to be perfect if M is closed and every point of M is a limit point of M.

Theorem 51: No countable point set is perfect.

Problem: Show "directly" that the unit interval is an uncountable set.

Theorem 52: Every region is uncountable.

Theorem 53: Every uncountable set has a limit point.

Definition: A point set H is said to be *nowhere dense* in a point set K if every region intersecting K contains a region which intersects K but not H.

Deny the above (i.e. state what would be meant by *not nowhere dense*). Use this for a definition of *dense at a point*.

Question: Can a set be perfect and nowhere dense?

Theorem 54: If H is nowhere dense in K, and T is a subset of H, then T is nowhere dense in K.

Theorem 55: If E_1 and E_2 are nowhere dense in M, then $E_1 + E_2$ is nowhere dense in M.

Question: Is there a 1-to-1 correspondence between the points of the unit interval and any uncountable subset of the unit interval?

Theorem 56: No region is the sum of a countable number of point sets that are nowhere dense in S.

Theorem 57: No closed point set M is the sum of a countable number of closed point sets such that if X is any one of them, every point of X is a limit point of $M - X$.

Theorem 58: No closed point set M is the sum of a countable number of point sets each nowhere dense in M.

Definition: A point set H is said to be everywhere dense in a point set K if $\overline{H}$ contains K.

Theorem 59: If the point set H is everywhere dense in the point set K, then every region intersecting K contains a point of H.

Theorem 59.5: There do not exist uncountably many mutually exclusive regions.

Theorem 60: If E is nowhere dense in the closed set M, then $M - E$ is everywhere dense in M.

Theorem 60.5: There exists an increasing unbounded sequence.

Question: Is the converse of Theorem 60 true?

Theorem 61: The intersection of a countable collection of open dense (in M) subsets of a closed point set M is dense in M.

Theorem 61.5: If p is a point, there exists an increasing sequence converging to p.

Definition: If the point set M contains a countable set which is everywhere dense in M, then M is said to be *separable*.

Axiom IV: S is separable.

Theorem 62: If p is a limit point of the point set M, there exists a sequence of points in M, there exists a sequence of points in M converging to p.

Theorem 63: There exists a countable collection F of regions such that if R is a region and p is a point of R, then some region of F contains p and lies in R.

Theorem 64: If G is a collection of regions covering the point set M, then some countable subcollection of G covers M.

Theorem 65: Every uncountable point set contains a limit point of itself.

Question: Is Theorem 63 equivalent to Axiom IV?

Theorem 66: Every point is separable.

Theorem 67: There does not exists a collection of mutually exclusive intervals covering space.

Theorem 68: Every closed uncountable set is the sum of a perfect set and a countable set.

Theorem 69: If M is an uncountable point set, then M is the sum of two sets H and K such that

(1) H is countable

(2) every region intersecting K contains uncountably many points of M.

Theorem 70: If M is a perfect set, then every region intersecting M contains uncountable many points of M.

Theorem 71: If every infinite subset of a point set M has a limit point in M, then M is closed and bounded (converse of 45).

Problem: Find a space which satisfies Axioms I, II, and III, but not IV.

Definition: A *transformation* f of a point set H onto a point set K is a collection of ordered pairs of points such that

(1) each ordered pair in f has a point of H as its first element and a point of K as its second element,

(2) every point of H is the first element of some ordered pair in f and every point of K is the second element of some ordered pair in f, and

(3) no two ordered pairs in f have the same first element.

Definitions and Notation: In the following definitions, let f be a transformation of a point set H onto a point set K.

If x is a point of H, then $f(x)$ denotes the second element of the ordered pair of f that has x as a first element.

If H' is a subset of H, then $f(H')$ denotes the set of all points $f(x)$ such that x is a point of H'.

If y is a point of H, then $f^{-1}(y)$ denotes the set of all points x in H such that $f(x) = y$.

If K' is a subset of K, then $f^{-1}(K')$ denotes the set of all points x in H such that $f(x)$ is a point of K'.

The transformation f is said to be one-to-one if no two ordered pairs in f have the same second element.

The transformation f is said to be continuous if for any sequence x_1, x_2, x_3, $\cdots$ of points in H converging to a point x in H; the sequence $f(x_1)$, $f(x_2)$, $f(x_3)$, $\cdots$ converges to $f(x)$.

If the transformation f is one-to-one, then f^{-1} denotes the collection of all ordered pairs $[y, f^{-1}(y)]$ such that y is a point of K.

If the transformation f is one-to-one and both f and f^{-1} are continuous, then f is said to be a homeomorphism (topological transformation) of H onto K, and H and K are said to be homeomorphic (topologically equivalent).

If K is a subset of K', the f is said to be a transformation of H into K'.

Theorem 72: If f is a continuous transformation of an interval H onto itself, then there is a point x of H such that $f(x) = x$.

Theorem 73: Let f be a continuous transformation of a point set H onto a point set K.

(1) If H is closed and bounded, then K is closed and bounded.

(2) If H is connected, then K is connected.

(3) If H is an interval, then K is either an interval or a point.

Definition: A transformation f of a space S_1 onto a space S_2 is said to *preserve order* if for any two points x and y in S_1 such that $x < y$, then $f(x) < f(y)$ in S_2.

Define "*reverse order*" similarly.

Theorem 74: If S_1 and S_2 are spaces satisfying Axioms 1, 2, 3, and 4, and f is an order preserving (reversing) transformation of S_1 onto S_2, then f is a homeomorphism.

Theorem 75: If S_1 and S_2 are spaces satisfying Axioms 1, 2, 3, and 4, and f is a homeomorphism of S_1 onto S_2, then f either preserves order or reverses order.

Theorem 76: Every two spaces satisfying Axioms 1, 2, 3, and 4 are homeomorphic.

Axiom 2′: S is non-degenerate and has a first point and a last point.

Note: It should be observed here that with the previous definition of a region, the first (last) point of S does not lie in a region. In addition to the sets previously defined as regions, consider the following sets as regions. For each point p different from the first (last) point of S, let the set of all points that precede (follow) p be a region.

Theorem 77: Every two spaces satisfying Axioms 1, 2′, 3, and 4 are homeomorphic.

Axiom 2″: S has a first point and no last point, or S has a last point and no first point.

Theorem 78: Every two spaces satisfying Axioms 1, 2″, 3, and 4 are homeomorphic.

